KERALA PUBLIC SERVICE COMMISSION • OFFICIAL SYLLABUS NOTE

SURVEYOR GRADE II (SURVEY & LAND RECORDS) & KWA TRACER

Cat. Nos: 411/2022, 692/2022, 106/2017, 043/2023 • Technical Education & Town Planning

MODULE 10

Building Materials and Construction

Official Weightage: 10 Marks Marks

PRESCRIBED SYLLABUS TOPICS COVERED:

Building Materials (Stones, Bricks, Cement, Mortar, Concrete, Timber, Steel), Brick Masonry Bonds (English vs Flemish, Closers), Foundations (Rankine's minimum depth), Roof Trusses (King Post & Queen Post), Basic Irrigation (Duty, Delta, Base Period 9CÓƒcD"ô@, Canals, Weirs & Barrages), Highway Engineering (Camber, Superelevation e=v²/225R, Sight Distances), Estimating & Costing (IS 1200 Units of Measurement, Center Line vs Long Wall-Short Wall Methods).

+P COMPREHENSIVE EXAM CRACKER FEATURES

- Strictly aligned with Kerala PSC latest question patterns (Land Records & KWA)
- Contains verified formulas, numerical shortcuts, standard tables & specifications
- Highlights repeatedly tested PSC trap questions & previous year keys

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CHAPTER 1: STONES & QUARRYING

Stones are natural building materials obtained from rocks by quarrying and dressing, widely used in masonry, foundations, dams, and road ballast.

1.1 Classification of Rocks

Classification Basis	Category	Geological Formation & Characteristics	Standard Examples
Geological	Igneous Rocks	Formed by cooling and solidification of molten magma or lava; crystalline, unstratified, high strength	Granite, Basalt, Trap, Pumice, Diorite
Geological	Sedimentary Rocks	Formed by deposition of weathered sediments in layers under water/wind; stratified, softer	Sandstone, Limestone, Shale, Conglomerate, Laterite
Geological	Metamorphic Rocks	Formed by transformation of igneous or sedimentary rocks under high heat and pressure	Marble (from Limestone), Slate (from Shale), Quartzite (from Sandstone), Gneiss (from Granite)
Physical	Stratified / Unstratified / Foliated	Stratified rocks have distinct bed planes (Limestone, Slate); Unstratified have no planes (Granite, Trap); Foliated split along definite direction (Gneiss, Schist)	Slate (Foliated), Granite (Unstratified)
Chemical	Siliceous / Argillaceous / Calcareous	Siliceous: Silica (SiO ,) main constituent (Granite, Quartzite); Argillaceous: Clay/ Alumina main (Slate, Laterite); Calcareous: Calcium carbonate CaCO ƒ main (Limestone, Marble)	Quartzite (Siliceous), Marble (Calcareous)

1.2 Tests on Stones & Mohs Hardness Scale

• **Crushing / Compressive Strength:** Tested on 40 mm or 50 mm stone cubes in a universal testing machine. Good building stones must have compressive strength greater than 100 N/mm² (1,000 kg/cm²). Granite has 100-140 N/mm²; Basalt has > 150 N/mm²; Sandstone has 65 N/mm².

• **Water Absorption Test:** Stone is soaked in clean water for 24 hours. A good building stone should NOT absorb water more than 5% of its dry weight. Stones absorbing > 10% water are rejected.

• **Mohs Scale of Mineral Hardness (1 to 10):** 1. Talc (Softest) -> 2. Gypsum -> 3. Calcite -> 4. Fluorite -> 5. Apatite -> 6. Feldspar -> 7. Quartz -> 8. Topaz -> 9. Corundum -> 10. Diamond (Hardest).

+P PSC FOCUS POINT: MOHS HARDNESS SCALE IN PSC

• 1 = Talc (Softest mineral, easily scratched by fingernail).
• 7 = Quartz.
• 10 = Diamond (Hardest mineral known).
• Fingernail hardness "H 2.5; Copper coin "H 3.5; Knife blade / Window glass "H 5.5.

CHAPTER 2: BRICKS & CLAY PRODUCTS

Bricks are artificial building blocks manufactured by moulding, drying, and burning suitable brick clay (earth).

2.1 Chemical Composition of Good Brick Earth

Constituent	Percentage	Function in Brick Manufacture	Defect / Hazard Caused if in Excess
Silica (Sand)	50% to 60%	Prevents cracking, shrinking, and warping of raw bricks; imparts uniform shape	Excess silica makes bricks brittle, weak, and causes lack of cohesion between particles
Alumina (Clay)	20% to 30%	Imparts plasticity to the earth so that it can be easily moulded without crumbling	Excess alumina causes raw bricks to shrink excessively and crack/ warp during drying and burning
Lime (CaO)	Less than 5%	Prevents shrinkage; acts as a flux during burning to fuse silica particles together	Excess lime causes brick to melt and lose its shape; lime lumps slake and split brick into pieces!
Iron Oxide (Fe ,O *f*)	5% to 6%	Imparts the characteristic pleasing RED color to burned brick; acts as a flux	Excess iron oxide turns bricks dark blue or blackish; deficiency results in pale yellow color
Magnesia (MgO)	Less than 1%	Imparts a yellowish tint; decreases shrinkage	Excess magnesia leads to decay of bricks in service

2.2 Harmful Ingredients in Brick Earth

- **Lime Lumps:** Quicklime absorbs moisture, slakes with volume expansion, and splits/disintegrates the brick.
- **Iron Pyrites:** Decomposes, oxidizes, and leaves black slag spots that discolour and crystallize out.
- **Alkalis (Soda & Potash):** Act as flux causing twisting. Leaches out to surface when exposed to rain, causing ugly white powdery salt patches called EFFLORESCENCE.
- **Pebbles & Gravel:** Prevent thorough mixing and cause weak cracking planes.
- **Organic / Vegetative Matter:** Burns away during firing, leaving porous hollow voids that reduce brick strength.

2.3 Standard Brick Sizes & Frog Specifications

- **Modular / Standard Size (IS 1077):** 190 mm × 90 mm × 90 mm (19 cm × 9 cm × 9 cm).
- **Nominal Size (With Mortar Joints):** 200 mm × 100 mm × 100 mm (20 cm × 10 cm × 10 cm). Standard mortar joint is 10 mm thick.
- **Traditional / Conventional Size:** 230 mm × 115 mm × 75 mm (9" × 4.5" × 3").
- **Number of Modular Bricks per m^3:** 1 m^3 / (0.2 m × 0.1 m × 0.1 m) = 500 BRICKS (Without mortar: approx 550 bricks including wastage).
- **Frog in Brick:** A depression (indentation) of 10 to 20 mm depth formed on the top face of a brick during moulding. Serves two purposes: (1) Acts as a shear key for mortar to bond consecutive courses, (2) Displays maker's trademark name.

2.4 Classification & Quality Tests of Bricks (IS 1077)

Brick Class	Minimum Compressive Strength	Maximum 24h Water Absorption	Standard Field Characteristics
First Class Brick	10.5 N/mm^2 (105 kg/cm^2)	20% of dry weight	Uniform deep red color; sharp rectangular edges; ringing metallic sound when struck; no efflorescence
Second Class Brick	7.0 N/mm^2 (70 kg/cm^2)	22% of dry weight	Slight surface irregularities allowed; clear ringing sound; used for internal walls and plastering
Third Class Brick	3.5 N/mm^2 (35 kg/cm^2)	25% of dry weight	Underburnt, light yellowish color; dull thud sound; used for temporary structures and road soiling

Brick Class	Minimum Compressive Strength	Maximum 24h Water Absorption	Standard Field Characteristics
Fourth Class (Overburnt)	High strength, brittle	Less than 5%	Overburnt vitrified bricks (Jhama); dark distorted; crushed into surkhi for lime concrete

+P PSC FOCUS POINT: BRICK COMPRESSIVE STRENGTH MINIMUMS

- First Class Brick: "e 10.5 N/mm² (Water absorption "d 20%).
- Second Class Brick: "e 7.0 N/mm² (Water absorption "d 22%).
- Common Building Brick (General Minimum): "e 3.5 N/mm² (Water absorption "d 25%).
- Standard Modular Size: 190 × 90 × 90 mm; Nominal size with mortar = 200 × 100 × 100 mm.

CHAPTER 3: CEMENT, SAND & MORTAR

Cement is the primary binding agent in modern construction, manufactured by calcining calcareous materials (limestone/ chalk) and argillaceous materials (clay/shale) at 1,400°C to 1,500°C into clinker.

3.1 Chemical Composition & Bogue's Compounds of Portland Cement

Bogue's Compound	Chemical Formula	Abbr.	Percentage	Hydration Characteristic & Structural Role
Tricalcium Silicate	3CaO · SiO ,	C ƒS	40% to 50%	Hydrates rapidly; responsible for EARLY STRENGTH (up to 7 to 28 days); high heat of hydration
Dicalcium Silicate	2CaO · SiO ,	C ,S	25% to 35%	Hydrates slowly over months; responsible for ULTIMATE LATER STRENGTH (after 1 year); low heat
Tricalcium Aluminate	3CaO · Al ,O ƒ	C ƒA	8% to 12%	Hydrates FIRST within 24 hours; causes INITIAL SET of cement; releases highest heat; vulnerable to sulphate attack
Tetracalcium Aluminoferrite	4CaO · Al ,O ƒ · Fe ,O ƒ	C „AF	6% to 10%	Poorest cementing value; hydrates rapidly; imparts grey color to cement

+P PSC FOCUS POINT: ROLE OF GYPSUM IN CEMENT

Raw clinker without gypsum sets instantaneously within minutes (flash set)! Raw gypsum (CaSO „·2H ,O, 2% to 3%) is added during final clinker grinding to RETARD the initial setting time of cement and allow adequate time for mixing and placing.

3.2 Standard Laboratory Tests on Cement

Test Name	Apparatus Used	Standard Test Procedure / Permissible Limit	PSC Exam Key
Standard Consistency	Vicat Apparatus (10 mm plunger)	Finds water percentage (P) required to make cement paste such that plunger penetrates to 5 to 7 mm from bottom of mould (usually 25% to 30%)	Plunger diameter = 10 mm; Mould height = 40 mm
Initial Setting Time	Vicat Apparatus (1 mm square needle)	Time from adding water until needle fails to pierce beyond 5 mm from bottom. NOT LESS THAN 30 MINUTES ("e 30 mins) for OPC	Tested with 0.85 P water
Final Setting Time	Vicat Apparatus (Annular circular collar with needle)	Time when needle makes an impression but annular collar fails to make an indent. NOT EXCEEDING 600 MINUTES ("d 10 hours) for OPC	Tested with 0.85 P water

Test Name	Apparatus Used	Standard Test Procedure / Permissible Limit	PSC Exam Key
Soundness Test	Le Chatelier Mould (for free lime) & Autoclave (for magnesia)	Measures expansion due to unburnt lime. Expansion of Le Chatelier indicator needles must NOT EXCEED 10 mm for OPC!	Autoclave limit "d 0.8%
Fineness Test	IS 90-micron sieve (Sieve No. 9)	Residue left on 90-micron sieve after 15 minutes of continuous sieving must NOT EXCEED 10% for Ordinary Portland Cement (OPC)	Specific surface area "e 2,250 cm²/g

3.3 Bulking of Sand

Bulking is the increase in volume of fine sand caused by surface tension films of moisture pushing sand grains apart when damp.

• **Moisture Content for Max Bulking:** Maximum bulking (up to 20% to 30% volume increase!) occurs at moisture content of 4% to 6% by weight.

• **Inundation Effect:** When sand is completely submerged in water (inundated), the surface tension films break down and the volume returns to that of completely dry sand!

• **Finest Sand Bulks Most:** Fine sand bulks significantly more than coarse sand because of its greater specific surface area.

CHAPTER 4: CONCRETE & REINFORCED CEMENT CONCRETE (RCC)

Concrete is an artificial stone-like composite material produced by mixing cement, fine aggregate (sand), coarse aggregate (gravel/crushed stone), and water in prescribed proportions.

4.1 Water-Cement Ratio & Abrams' Law

Abrams' Law states that for fully compacted concrete at a given age, the compressive strength depends purely on the ratio of water to cement by weight (Water-Cement Ratio, w/c), independent of aggregate proportions!

Ø=ÜÐ CORE FORMULA VAULT: WATER-CEMENT RATIO PRINCIPLE

Compressive Strength = K • / (K ,^(w/c))

Lower Water-Cement Ratio => Higher Compressive Strength & Durability!

Higher Water-Cement Ratio => Higher Workability, BUT severely reduced strength and porous concrete.

Standard w/c ratio for normal structural concrete is 0.40 to 0.55.

4.2 Workability Tests & Standard Concrete Grades

Concrete Grade (IS 456)	Nominal Volumetric Mix Ratio (Cement : Sand : Coarse Agg)	Characteristic 28-Day Strength (fck on 150 mm	Standard Construction Application
M10	1 : 3 : 6	10 N/mm² (100 kg/cm²)	Plain cement concrete (PCC) foundations, levelling course
M15	1 : 2 : 4	15 N/mm² (150 kg/cm²)	General mass concrete, non-structural pavements
M20	1 : 1.5 : 3	20 N/mm² (200 kg/cm²)	MINIMUM GRADE for Reinforced Cement Concrete (RCC) in normal conditions!
M25	1 : 1 : 2	25 N/mm² (250 kg/cm²)	Heavy RCC columns, beams, retaining walls, cantilever slabs

+P PSC FOCUS POINT: MINIMUM GRADE OF RCC (IS 456:2000)

Under IS 456:2000, the MINIMUM permissible grade of concrete for structural Reinforced Cement Concrete (RCC) is M20 (1:1.5:3). For prestressed concrete, minimum grade is M30 for post-tensioned and M40 for pre-tensioned!

CHAPTER 5: TIMBER, PAINTS & BUILDING TRUSSES

Wood from exogenous trees processed for structural use is called timber.

5.1 Cross-Sectional Structure of an Exogenous Tree Trunk

Layer (Inside to Outside)	Description & Anatomical Structure	Engineering Importance
1. Pith (Medulla)	Innermost central core of dark, soft cellular tissue	Oldest part of tree; useless for structural timber; easily rots
2. Heartwood (Duramen)	Inner annular rings surrounding the pith; dark, dense, dead wood	STRONGEST and most durable portion; ideal for building timber!
3. Sapwood (Alburnum)	Outer light-colored rings between heartwood and cambium; active living sap cells	Soft, weak, prone to fungal decay and insect attack; inferior timber
4. Cambium Layer	Thin microscopic layer of sap between sapwood and inner bark	Active growing layer of living cells forming new wood rings annually
5. Inner Bark (Bast)	Fibrous inner layer protecting cambium	Transports organic nutrients down the tree
6. Outer Bark (Cortex)	Tough outermost protective skin of tree	Shields trunk from weather and mechanical injury
7. Medullary Rays	Thin radial cellular lines running from pith outwards to bark	Holds annular rings together; imparts radial strength

5.2 Roof Trusses: King Post vs Queen Post

Truss Type	Economical Span Range	Number of Vertical Tension Posts	PSC Structural Distinctions
King Post Truss	5 to 8 metres span (up to 9 m)	ONE central vertical tension post (The King Post)	King post connects the apex of the rafters to the central tie beam in pure tension!
Queen Post Truss	8 to 12 metres span (up to 14 m)	TWO vertical tension posts (The Queen Posts)	Two queen posts connected at top by a horizontal straining beam and at bottom by tie beam

CHAPTER 6: FOUNDATIONS & BRICK MASONRY BONDS

Foundations transmit all dead, live, and wind loads from the superstructure safely to the subsoil without exceeding the safe bearing capacity.

6.1 Rankine's Minimum Depth of Foundation Formula

Ø=ÜÐ CORE FORMULA VAULT: RANKINE'S FOUNDATION DEPTH FORMULA

Minimum Depth of Foundation d_min = (p / w) × [(1 - sin <b’ ò ƒ ² 6–â Æ)]²

where p = safe bearing capacity of soil (kN/m²), w = unit weight of soil (kN/m³), <b Ò ævÆR öb –çFW&æ Â g&–7F–öâ öb soil

Notice: For cohesive clays where <b Ò À Rankine's formula gives d = p / w.

6.2 Brick Masonry Bonds: English vs Flemish Bond

Bond Type	Arrangement of Brick Courses	Strength vs Appearance	Standard Features
English Bond	Alternate courses of ALL HEADERS and ALL STRETCHERS throughout the wall	STRONGEST brick masonry bond; maximum structural load-bearing capacity	Queen closer placed immediately after the corner quoin header to create 1/2 brick lap
Flemish Bond	Each course contains ALTERNATE HEADERS AND STRETCHERS along the same course	Better, more artistic and decorative appearance; slightly weaker than English bond	Double Flemish shows Flemish pattern on both faces; Single Flemish shows Flemish on front and English on back

+P PSC FOCUS POINT: BRICK CLOSERS IN MASONRY

• Queen Closer: A brick cut LENGTHWISE into two equal halves (width = 45 mm, length = 190 mm). Placed next to the quoin header to break vertical joints.

• King Closer: Formed by cutting off a triangular corner such that half-header and half-stretcher remain (cut across midpoints of adjacent sides).

CHAPTER 7: BASIC IRRIGATION & WATER AUTHORITY TRACER CONCEPTS

Irrigation engineering concepts frequently tested in KWA Tracer and Surveyor Grade II examinations.

7.1 Duty, Delta & Base Period

Irrigation Term	Symbol	Unit of Measurement	Rigorous Definition & Engineering Relationship
Duty	D	Hectares per cumec (ha/cumec)	The total area of land that can be irrigated by a continuous discharge of 1 cubic metre per second (cumec) of water throughout the crop base period
Delta	9@	Centimetres (cm) or metres (m)	The total accumulated depth of water required by a crop during its entire life cycle from sowing to harvest
Base Period	B	Days	The period of time (in days) from the very first watering of a crop at sowing to its final watering just before harvesting

Ø=ÜÐ CORE FORMULA VAULT: DUTY-DELTA-BASE PERIOD RELATIONSHIP

Delta 9B Ò ƒƒcB r "' ò B ¶–â 6VçF–ÖWG&W5Ð

Delta 9B Ò ƒ,ãcB r "' ò B ¶–â ÖWG&W5Ð

where B = Base period in days, D = Duty in hectares/cumec

Rule: Higher Duty indicates higher irrigation efficiency (less water required per hectare)!

7.2 Canal Cross-Section & Hydraulic Structures

• **Berm:** The horizontal strip of land left between the toe of the canal bank and the top edge of the cutting. Functions: (1) Prevents bank sliding into canal, (2) Allows silt deposition to form protective lining, (3) Provides wide path for maintenance inspection.

• **Freeboard:** The vertical distance between the full supply water level (FSL) and the top of the canal bank. Prevents waves from overtopping the bank (typically 0.5 m to 0.75 m).

• **Weir vs Barrage:** Weir: A low masonry obstruction across a river with a raised permanent crest wall; water level is

maintained mostly by the permanent crest. Barrage: Has practically no crest wall; water level and afflux are regulated ENTIRELY by adjustable mechanical gates!

CHAPTER 8: HIGHWAY CROSS-SECTION & GEOMETRIC DESIGN

Road geometry design principles governing civil works.

8.1 Camber (Cross-Fall) & Superelevation

Highway Element	Purpose	IRC Standard Values for Plain Terrain
Camber (Cross-Fall)	Transverse slope provided to the road surface to drain off rainwater rapidly into side drains	Concrete / Heavy Bitumen: 1 in 50 (2%) to 1 in 60 (1.7%) Thin Bitumen: 1 in 40 (2.5%) to 1 in 50 (2%) Water Bound Macadam (WBM): 1 in 33 (3%) to 1 in 40 (2.5%) Earthen roads: 1 in 25 (4%) to 1 in 33 (3%)
Carriageway Width	Paved width allocated strictly for vehicular traffic movement	Single Lane: 3.75 m Two-lane without raised kerbs: 7.0 m Two-lane with raised kerbs: 7.5 m Multi-lane: 3.5 m per lane
Superelevation (e)	Transverse tilt provided on horizontal curves to counteract centrifugal force	Design formula for mixed traffic (75% design speed): $e = v^2 / (225 \cdot R)$ Maximum permissible e = 7% (0.07) for plain terrain; 10% (0.10) for hilly roads

CHAPTER 9: BASIC ESTIMATING & COSTING (IS 1200)

Estimating determines the probable quantity of materials, labor, and cost before commencing construction.

9.1 Standard Units of Measurement (IS 1200)

Civil Engineering Work	Unit of Measurement	Unit of Payment	Measurement Standard Notes
Earthwork excavation in foundation	Cubic Metre (m^3)	Per 100 m^3 or m^3	Lead = 50 m; Lift = 1.5 m standard
Concrete in foundation & RCC roof	Cubic Metre (m^3)	Per m^3	Formwork measured separately in m^2
Damp Proof Course (DPC)	Square Metre (m^2)	Per m^2	Thickness specified in description (usually 2.5 cm)
Brickwork in wall (> 10 cm thick)	Cubic Metre (m^3)	Per m^3	Half-brick partition walls measured in m^2
Plastering, Pointing, Painting	Square Metre (m^2)	Per m^2	Deductions for openings as per IS 1200
Steel Reinforcement in RCC	Quintal (q) or kg	Per Quintal (100 kg)	Includes cutting, bending, and placing

9.2 Methods of Building Estimates

• **Center Line Method:** The total centerline length of all continuous walls is calculated. For T-junctions where partition walls join main walls, a deduction of HALF THE WALL THICKNESS (1/2 t) is made for every T-junction. Net Centerline = Total Centerline - (Number of T-junctions × 1/2 t). Very fast and accurate for symmetrical plans!

• **Long Wall - Short Wall Method:** Long walls are measured out-to-out: Length = In-to-in distance + 2 × (1/2 wall thickness) = In-to-in + wall thickness. Short walls are measured in-to-in: Length = In-to-in distance - 2 × (1/2 wall thickness). Length of long wall decreases as we move from foundation to superstructure; length of short wall increases!

CHAPTER 10: PSC SOLVED NUMERICAL MODELS

Ø=ÜÝ PSC SOLVED NUMERICAL: Duty-Delta-Base Period Calculation

Question: A crop with a base period of 120 days requires a total depth of water (delta) of 96 cm. Find the duty of canal water in hectares per cumec.

Solution:
Formula: 9B Ò ƒ ƒcB r '" ò B ‡v†W&R " is in cm, B in days, D in ha/cumec)
96 = (864 × 120) / D
D = (864 × 120) / 96 = 9 × 120 = 1,080 hectares/cumec

Correct Option / Answer: 1,080 hectares/cumec

Ø=ÜÝ PSC SOLVED NUMERICAL: Rankine's Minimum Depth of Foundation

Question: Determine the minimum depth of foundation for a building column bearing on soil with a safe bearing capacity of 150 kN/m², unit weight of soil = 18 kN/m³, and angle of internal friction <b Ò 3 à

Solution:
Given: p = 150 kN/m², w = 18 kN/m³, <b Ò 3
sin 30° = 0.5 => (1 - sin 30°) / (1 + sin 30°) = (1 - 0.5) / (1 + 0.5) = 0.5 / 1.5 = 1 / 3
Rankine's formula: d_min = (p / w) × [(1 - sin <b' ò ƒ ² 6–â Æ)]²
d_min = (150 / 18) × (1 / 3)² = (8.333) × (1 / 9) = 0.926 m "H 0.93 m

Correct Option / Answer: 0.93 metres

Ø=ÜÝ PSC SOLVED NUMERICAL: Superelevation on Horizontal Road Curve

Question: Calculate the required superelevation for a mixed traffic highway on a horizontal curve of radius 300 m for a design speed of 60 km/h.

Solution:
Formula (IRC mixed traffic): e = v² / (225 × R)
Speed v = 60 km/h, Radius R = 300 m
e = (60)² / (225 × 300) = 3,600 / 67,500 = 0.0533 = 5.33%
Check: 5.33% is less than the maximum allowable limit of 7% (0.07). Hence, e = 0.0533 is safe.

Correct Option / Answer: 0.0533 (5.33% or 1 in 18.75)

CHAPTER 11: KERALA PSC 50 RAPID-FIRE ONE-LINERS & EXAM VAULT

Rapid-Fire PSC One-Liners: Building Materials

- **Chief ingredient of good brick earth:** Silica (50% to 60%).
- **Ingredient imparting plasticity to brick earth:** Alumina (20% to 30%).
- **Standard modular brick size as per BIS:** 190 × 90 × 90 mm.
- **Nominal modular brick size with mortar:** 200 × 100 × 100 mm.
- **Number of modular bricks in 1 cubic metre:** 500 bricks.
- **Minimum compressive strength of 1st class brick:** 10.5 N/mm² (Water absorption "d 20%).
- **Component added to clinker to retard setting time:** Gypsum (2% to 3%).
- **Apparatus used to test initial and final setting time:** Vicat Apparatus.
- **Initial setting time of Ordinary Portland Cement:** Not less than 30 minutes ("e 30 mins).
- **Final setting time of Ordinary Portland Cement:** Not exceeding 600 minutes ("d 10 hours).

Rapid-Fire PSC One-Liners: Concrete, Timber & Masonry

- **Minimum grade of concrete for structural RCC:** M20 (1 : 1.5 : 3).
- **Law stating concrete strength depends on w/c ratio:** Abrams' Law.
- **Moisture content of sand causing maximum bulking:** 4% to 6% moisture content.

- **Dark, dense, strongest inner wood of tree trunk:** Heartwood (Duramen).
- **Strongest bond in brick masonry:** English Bond.
- **Brick cut lengthwise into two equal halves:** Queen Closer.
- **Roof truss suitable for spans up to 8 metres:** King Post Truss (single vertical tie).
- **Roof truss suitable for spans of 8 to 12 metres:** Queen Post Truss (two vertical ties).
- **Unit of measurement for Damp Proof Course (DPC):** Square Metre (m^2).
- **In center line method, deduction at each T-junction is:** Half the wall thickness (1/2 t).

Rapid-Fire PSC One-Liners: Irrigation & Highways

- **Fundamental equation connecting Delta, Duty, Base period:** 9B Ò ƒ ƒcB r ‴ ò B †6Ò’à
- **Transverse road slope to drain off surface water:** Camber (Cross-fall).
- **Standard carriageway width for single lane road:** 3.75 metres.
- **Superelevation design formula for mixed traffic (IRC):** $e = v^2 / (225 \cdot R)$.
- **Maximum permissible superelevation on plain roads:** 7% (0.07).
- **Horizontal ledge left between canal cutting and bank:** Berm.
- **Hydraulic structure regulating water level purely by gates:** Barrage.
- **Crop period from sowing to harvest is called:** Base Period (in days).
- **Mineral ranking 1 on Mohs scale of hardness:** Talc (Softest).
- **Mineral ranking 10 on Mohs scale of hardness:** Diamond (Hardest).